A degree-separated equation in nine unknowns

Explicit Diophantine consistency reduction

23 October 2026

Abstract

A new degree-separated source code combines with sparse carry padding, an even-parameter parity calibration, and a positivity-first Pell finalizer to produce an explicit integer polynomial in nine positive unknowns of exact degree 2,676,548. Its supplied circuit uses 476 binary arithmetic operations. Against the previous degree 7,344,328 and 738-operation construction, these are reductions of 63.56% and 35.50%. All fixed coefficient payloads, finite certificates, and authored rebuilding and checking programs are supplied. Absence of zeros is provably equivalent in ZFC to Con(ZFC), using the identified inherited compiler and Jones ratio theorems. No proof of consistency or global minimality is claimed.

1 A degree-separated nine-variable equation

The primary equation has nine strictly positive unknowns, exact total degree 2 676 548, and 476 live binary operations. Its separately translated nonnegative form has the same degree. The complete division-free DAGs, coefficient payloads, exact-degree certificates, and a standard-library checker accompany this section. Register names below abbreviate polynomials; they introduce no additional unknowns.

Scope. The ordinary mathematical theorem is that ZFC proves existence of a zero in the stated domain equivalent to existence of a finite syntactic ZFC refutation. Its negation is the requested consistency equivalence. The original package31 compiler and the numerical Jones ratio theorems are explicit imported dependencies. This is not a proof of consistency, a new end-to-end kernel formalization, an unrestricted-integer nine-variable claim, or a global minimum claim.

1.1 A smaller source by keeping the three degrees separate

Use the original 31-positive-variable, 22-residual source $r_1, \dots, r_{22}$, with its *whole* original coefficient tuple. The four payload bit lengths are $1064921, 1070004, 1064921, 1064922$; the fourth port is already shifted. Replace just

$$\rho_{14} = r_{14} + r_2, \qquad \rho_{15} = r_{15} + r_3.$$

The inverse row operations recover the old system, so its zero set is unchanged. It has 89 weighted records: 7 constant, 39 linear, and 43 quadratic.

For positive source coordinates y_i, write $U_y(t) = \sum_i y_i t^{w_i}$, with weights

$$(1, 7, 61, 36, 16, 85, 20, 67, 43, 11, 34, 13, 96, 80, 29, 50,\\ 2, 109, 110, 112, 116, 91, 53, 55, 27, 101, 92, 9, 107, 37, 60).$$

The maximum weight is $W = 116$. For a monomial of degree d, weight its coefficient by 2 when $d = 0, 1$, and by $\prod_i \alpha_i!$ when $d = 2$. With the fixed target assigned to its row, form

$$D_d(t) = \sum_{j, |\alpha| = d} c_{j,\alpha} t^{q_j - \sum_i \alpha_i w_i}, \qquad \mathcal{G}(t, y) = D_2(t) U_y(t)^2 + D_1(t) U_y(t) + D_0(t).$$

The finite target certificate makes a degree-d term hit a marked target only against its own monomial of that same degree. Cross-degree label collisions are harmless, because the three kernels are separate summands. Checking all 22 544 interactions proves the formal identities

$$[t^{q_j}]\mathcal{G}(t,y) = 2\rho_j(y).$$

The kernel degrees are $(1719, 2027, 1929)$, the leading coefficient of D_2 is $+1$, and the convolution height is $K_{\mathrm{code}} = 2161$. The largest target is 2129. All 89 kernel records, coefficient recipes, and the row-target map are supplied as exact JSON.

1.2 Positive words and unconditional source bounds

Every weighted coefficient has absolute value below $M = 2^{2134927}$. Set the fixed ordinary integer coefficients

$$a = 4096M = 2^{2134939}, \qquad c = 2^{17079502}.$$

Use mixed coordinates $b, g, h, s, w, i \geq 1$ and $\phi, j, n \geq 0$. Define

$$\begin{aligned}
&B = ab^2, && \ell = \sum_i B^{w_i}, && U_s = \ell + g,\\
&G_s = (D_2(B)U_s + D_1(B))U_s + D_0(B), && A_q = \sum_j B^{q_j}, &&\\
&N_0 = B^{117}, && Z = B^{2162}, && M_c = N_0 - 1 - (b-1)\ell,\\
&J_c = 2G_s + A_q, && S_c = g + N_0 J_c, && T_c = M_c + N_0(B-2)A_q,\\
&L = 2bB^{2279}, && N = cbB^{1201}, && G_{\mathrm{cap}} = 32bB^{116},\\
&R = L(L + S_c + T_c + 1) - S_c - 1. &&&&
\end{aligned}$$

Here L is the binary-block length, while N is the separate square-root threshold for Jones's theorem.

Positivity must precede digit recovery. For every $g \geq 1$, $U_s \geq B^{116}$, and geometric bounds give

$$D_2(B) > \tfrac{1}{2}B^{1929}, \quad |D_1(B)| < 2MB^{2027}, \quad |D_0(B)| < 2MB^{1719}.$$

Relative to $B^{1929}U_s^2$, the negative contributions are bounded by $2M/B^{18}$ and $2M/B^{442}$, each below $1/8$. Thus $G_s > \frac{1}{4}B^{1929}U_s^2 > 0$. Also $0 \leq M_c < N_0$, and

$$R = L^2 + (L-1)S_c + LT_c + L - 1 \geq L^2 + L - 1.$$

Consequently $R, N \geq 8$, $b \leq R, N$, and $G_{\mathrm{cap}} > 0$ on every outer assignment, without first assuming the final equation vanishes.

On the branch $g < 4bB^{116}$, one has $U_s < 6bB^{116}$, and

$$|G_s| < \frac{43Mb^2B^{2162}}{B-1}, \qquad 2G_s < Z/32, \qquad A_q, (B-2)A_q < Z/64.$$

Hence $g < N_0$, $J_c < 3Z/64$, and

$$S_c + T_c < N_0 Z = B^{2279} < L.$$

These are numerical, prebinary bounds, not circular consequences of the carry test.

If b is a power of two, so is B. The low-mask test $g \,\&\, M_c = 0$ says exactly $g = \sum_i z_i B^{w_i}$, $0 \leq z_i < b$. Then $y_i = z_i + 1$ are positive and $U_s = \sum_i y_i B^{w_i}$. Each coefficient h_t of $\mathcal{G}(t,y)$, including unsupported collisions, satisfies

$$|h_t| < M\big((\textstyle\sum_i y_i)^2 + \sum_i y_i + 1\big) \leq 993Mb^2 < B/4.$$

The signed digits of J_c are $2h_t+1$ at targets and $2h_t$ elsewhere. Normalization has incoming and outgoing borrows in $\{-1,0\}$. At a target, the normalized digit avoids $B-2$ exactly when $h_t=0$: the allowed digits are 0 or 1, while positive or negative h_t activate a masked bit. Therefore

$$S_c \,\&\, T_c = 0 \quad \Longleftrightarrow \quad g \text{ encodes a positive common zero of the source.}$$

Conversely, any source zero admits every sufficiently large power-of-two b; its digits are $z_i = y_i - 1$. The all-one source tuple is not a zero, because $r_5(1,\dots,1)=1$. Thus $g>0$, and $g < bB^{116}$, giving the stronger completeness bound $32g < G_{\mathrm{cap}}$.

1.3 A calibrated two-block threshold

For $S,T \ge 0$, $S+T<L=2^u$, the integer

$$X = L^2 + (S+T)L + (L-1-S)$$

has nonoverlapping binary blocks. Consequently

$$\mathrm{pop}(X) = u + \mathrm{pop}(T) + 1 - \big[\mathrm{pop}(S) + \mathrm{pop}(T) - \mathrm{pop}(S+T)\big].$$

The bracket is nonnegative and vanishes exactly for carry-free addition. Legendre's formula identifies $\mathrm{pop}(X)$ with $v_2\binom{2X}{X}$.

Write $b=2^k$, $B=2^v$, $v=2134939+2k$. The unchanged low mask has $117v-31k$ set bits and the 22 target cells contribute $22(v-1)$. Hence

$$\mathrm{pop}(T_c) = 139v - 31k - 22, \qquad u = 1 + k + 2279v.$$

The exact threshold is

$$u + \mathrm{pop}(T_c) + 1 = 2418v - 30k - 20 = 2(17079502 + k + 1201v) = 2\log_2 N.$$

The factor b in N accommodates the even parameter 2418, without restricting b to a sparse family. We have proved

$$N^2 \mid \binom{2R}{R} \quad \Longleftrightarrow \quad S_c \,\&\, T_c = 0.$$

1.4 Ratio registers and the positivity-first finalizer

Define

$$\begin{aligned}
U &= N^2 w, & Y &= N^2 s, & M_0 &= RY,\\
P &= 2M_0^2 U, & K &= R+1+h(P-1), & A &= M_0(U+1),\\
B_1 &= 2R+1, & W_1 &= bw, & C &= B_1 + W_1 + \phi,\\
J &= 4A-5, & Q &= 3W_1 C - 2(W_1^2-1), & &\\
D &= (A^2-1)C^2+1, & k_0 &= iJC, & E &= k_0 DC,\\
F &= (A^2-1)E^2+1, & G &= A + F(CD+1-A), & &\\
H &= C + F(B_1 + jC), & I &= (G^2-1)H^2+1, & &\\
a_0 &= (P^2-1)K^2+1, & a_1 &= DFI. & &
\end{aligned}$$

Since $F \equiv 1 \pmod J$ and $F \equiv 1 \pmod C$, the old divisibility $JF \mid QF + J(H_{\mathrm{old}} - C)$ is equivalent to $J \mid Q$ and $F \mid H_{\mathrm{old}} - C$. The exact witness substitution

$$H = C + F(B_1 + jC), \qquad j_{\mathrm{old}} = 1 + \frac{F-1}{C}B_1 + Fj \ge 1$$

parametrizes precisely those old witnesses; the quotient is an integer. It introduces no new variable.

Use the polynomial margin with its known factor 8 removed:

$$t = bB^{116}, \qquad m = K^2(4t - g) - 16t(C - KY)^2.$$

Then $8m$ is exactly the original Jones margin. Its strict sign is unchanged. The imported ratio interface is now

$$a_0 \text{ square}, \quad a_1 \text{ square}, \quad J \mid Q, \quad m > 0.$$

Soundness gives b a power of two, $8g < G_{\mathrm{cap}}$, and the central-binomial divisibility. Completeness uses $32g < G_{\mathrm{cap}}$. All of its numerical hypotheses were established before this invocation.

The strict positive majorants

$$V_0 = PK, \qquad T_* = k_0(8D^2 - 4D - 1), \qquad V_1 = T_*GH$$

satisfy

$$\begin{aligned} V_0^2 - a_0 &= K^2 - 1 > 0, \\ T_*^2 - 64DF &= (8D+1)k_0^2 - 64D > 0, \\ \Delta_1 := V_1^2 - 64a_1 &= \big((8D+1)k_0^2 - 64D\big)(GH)^2 + 64DF(H^2 - 1) > 0. \end{aligned}$$

Also $0 < Q/J < Q < 3C^2 < 3D < T_* \le V_1$. The promised equation is

$$\Omega = Jm, \qquad \Theta = Jn + Q - \Omega(V_0 + V_1).$$

$$\boxed{F_9 = \big[\Theta^2 - \Omega^2(a_0 + 64a_1)\big]^2 - 256\Omega^4 a_0 a_1 = 0.}$$

If $m = 0$, then $\Theta = Jn + Q > 0$, so there is no zero. Otherwise the real norm factors give roots $r_0^2 = a_0$, $r_1^2 = 64a_1$ and

$$n + Q/J = m(V_0 + V_1 + r_0 + r_1).$$

The majorants make the parenthesis positive before any integrality claim. Thus $m > 0$. The margin gives $C > K$, and the elementary inequalities $A^2 > P$, $D > C^2$, $F \ge A^2$, $G \ge A$, $I \ge G^2$ give $a_1 > a_0$. Hence the rational signed sum $x = \Theta/\Omega = r_0 + r_1$ is nonzero. From $2xr_0 = x^2 + a_0 - 64a_1$, both roots are rational and therefore integral; $r_1/8$ is integral as well. The factor equation then yields $J \mid Q$. Conversely choose positive roots and set $n = m(V_0 + V_1 + r_0 + r_1) - Q/J > 0$. This proves the finalizer equivalence.

1.5 Exact degree and implementation certificate

The diagonal degrees are $r = \deg R = 9118$, $t = \deg N = 2403$. The degree-certificate circuit resolves the near-square cancellation by

$$\begin{aligned} &\Theta^2 - \Omega^2(a_0 + 64a_1) \\ &\quad = E_0^2 - 2E_0\Omega(V_0 + V_1) + \Omega^2\big[(K^2 - 1) + \Delta_1 + 2V_0V_1\big], \qquad E_0 = Jn + Q. \end{aligned}$$

The unique dominant term is $\Omega^2\Delta_1$, of degree 1,338,274. Its square dominates the final subtracted norm term, of degree 1,823,794. Thus

$$\deg F_9 = 188r + 400t + 8W + 236 = 2676548, \qquad \mathrm{lc}_{\mathrm{diag}}(F_9) = 2^{542}a^{1337768}c^{400} > 0.$$

The diagonal attains the full multivariate upper bound, proving equality. The positive coordinates replace ϕ, j, n by their positive inputs minus 1; the natural version instead increments the other six inputs. These are domain bijections, not integer-square parametrizations.

The supplied checker authenticates both separate whole tuples, rebuilds all 16 primary/comparison and certificate circuits, proves 24 source-evaluator polynomial identities, and replays 480 actual-tuple modular assignments. It additionally exercises 43,945 binary-block cases, 540 threshold identities, Pell-majorant and CRT identities, and finalizer completeness. These finite checks audit the implementation; they do not replace the universal proof. The source and full theorem are detailed in `nine/PROOF.md`.